

Integration Manual

for S32G FR_LLCE Driver

Document Number: IM11FRLLCEASR4.4 Rev0000R1.0.7 Rev. 1.0

1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	4
2.3 About This Manual	4
2.4 Acronyms and Definitions	5
2.5 Reference List	5
3 Building the driver	6
3.1 Build Options	6
3.1.1 GCC Compiler/Assembler/Linker Options	6
3.1.2 GHS Compiler/Assembler/Linker Options	9
3.2 Files required for compilation	11
3.3 Setting up the plugins	13
4 Function calls to module	15
4.1 Function Calls during Start-up	15
4.2 Function Calls during Shutdown	15
4.3 Function Calls during Wake-up	15
5 Module requirements	16
5.1 Exclusive areas to be defined in BSW scheduler	16
5.2 Peripheral Hardware Requirements	21
5.3 ISR to Configure within OS - Dependencies	21
5.4 ISR Macro	22
5.4.1 Without an Operating System	22
5.4.2 With an Operating System	22
5.5 Other AUTOSAR modules - dependencies	22
5.6 Data Cache Restrictions	23
5.7 User Mode support	23
5.7.1 User Mode configuration in the module	23
5.7.2 User Mode configuration in AutosarOS	24
5.8 Multicore support	25
6 Main API Requirements	26
6.1 Main function calls within BSW scheduler	26
6.2 API Requirements	26
6.3 Calls to Notification Functions, Callbacks, Callouts	26
7 Memory allocation	27
7.1 Sections to be defined in Fr_43_LLCE_MemMap.h	27



7.2 Linker command file	28
8 Integration Steps	29
9 External assumptions for driver	30



Chapter 1

Revision History

Revision	Date	Author	Description
1.0	29.09.2023	NXP LLCE Team	Prepared for release S32G LLCE AUTOSAR 4.4 Version 1.0.7

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for Flexray Driver for S32 microcontrollers.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of Flexray Driver	AUTOSAR Release 4.4.0
2	S32G2 Reference Manual	Rev. 5, 05/2022
3	S32G2 Data Sheet	Rev. 5, 05/2022
4	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 0P77B, Rev. 3.1
5	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 1P77B, Rev. 1.1
6	S32G3 Reference Manual	Rev. 2, 09/2022
7	S32G3 Data Sheet	Rev. 2, Draft C, 09/2021
8	S32G3 Errata Document	S32G3 Mask Set Errata for Mask 1P72B, Rev. 2.2

Chapter 3

Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 9.2.0 20190812 (Build 1649 Revision gaf57174) (for both S32G2 and S32G3)
- Green Hills Multi 7.1.6d / Compiler 2020.1.4 (for both S32G2 and S32G3)

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS_T40D11M10I7R0 part of the plugin name is composed as follows:

- T = Target_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative_Id (e.g. D11 identifies S32 platform)
- M = SW_Version_Major and SW_Version_Minor
- I = SW_Version_Patch
- R = Reserved

3.1.1 GCC Compiler/Assembler/Linker Options

3.1.1.1 GCC Compiler Options

Compiler Option	Description
-mcpu=cortex-m7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mlittle-endian	Generate code for a processor running in little-endian mode
-mfpv=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-std=c99	Specifies the ISO C99 base standard
-Os	Optimize for size. Enables all -O2 optimizations except those that often increase code size
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-Wall	Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros
-Wextra	This enables some extra warning flags that are not enabled by -Wall
-pedantic	Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option
-Wstrict-prototypes	Warn if a function is declared or defined without specifying the argument types
-Wundef	Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero
-Wunused	Warn whenever a function, variable, label, value, macro is unused
-Werror=implicit-function-declaration	Make the specified warning into an error. This option throws an error when a function is used before being declared
-Wsign-compare	Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.
-Wdouble-promotion	Give a warning when a value of type float is implicitly promoted to double
-fno-short-enums	Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.
-funsigned-char	Let the type char be unsigned by default, when the declaration does not use either signed or unsigned
-funsigned-bitfields	Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned
-fomit-frame-pointer	Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.

Compiler Option	Description
-fno-common	Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit
-fstack-usage	This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis
-fdump-ipa-all	This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps
-c	Stop after assembly and produce an object file for each source file
-DS32XX	Predefine S32XX as a macro, with definition 1
-DS32G2XX	Predefine S32G2XX as a macro, with definition 1
-DS32G3XX	Predefine S32G3XX as a macro, with definition 1
-DS32R45	Predefine S32R45 as a macro, with definition 1
-DGCC	Predefine GCC as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↔ c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT↔ RT as a macro, with definition 1. Allows drivers to be configured in user mode.

3.1.1.2 GCC Assembler Options

Assembler Option	Description
-X assembler-with-cpp	Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)
-mcpu=cortexm7	Targeted ARM processor for which GCC should tune the performance of the code
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mthumb	Generates code that executes in Thumb state
-c	Stop after assembly and produce an object file for each source file

3.1.1.3 GCC Linker Options

Linker Option	Description
-Wl,-Map,filename	Produces a map file
-T linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
-entry=Reset_Handler	Specifies that the program entry point is Reset_Handler
-nostartfiles	Do not use the standard system startup files when linking
-mcpu=cortexm7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mlittle-endian	Generate code for a processor running in little-endian mode
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-lc	Link with the C library
-lm	Link with the Math library
-lgcc	Link with the GCC library

3.1.2 GHS Compiler/Assembler/Linker Options

3.1.2.1 GHS Compiler Options

Compiler Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-thumb	Selects generating code that executes in Thumb state
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-C99	Use (strict ISO) C99 standard (without extensions)
-ghstd=last	Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)
-Osize	Optimize for size
-gnu_asm	Enables GNU extended asm syntax support
-dual_debug	Generate DWARF 2.0 debug information
-G	Generate debug information
-keeptempfiles	Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory

Compiler Option	Description
-Wimplicit-int	Produce warnings if functions are assumed to return int
-Wshadow	Produce warnings if variables are shadowed
-Wtrigraphs	Produce warnings if trigraphs are detected
-Wundef	Produce a warning if undefined identifiers are used in #if preprocessor statements
-unsigned_chars	Let the type char be unsigned, like unsigned char
-unsigned_fields	Bitfields declared with an integer type are unsigned
-no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup
-no_exceptions	Disables C++ support for exception handling
-no_slash_comment	C++ style // comments are not accepted and generate errors
-prototype_errors	Controls the treatment of functions referenced or called when no prototype has been provided
-incorrect_pragma_warnings	Controls the treatment of valid #pragma directives that use the wrong syntax
-c	Stop after assembly and produce an object file for each source file
-DS32XX	Predefine S32XX as a macro, with definition 1
-DS32G2XX	Predefine S32G2XX as a macro, with definition 1
-DS32G3XX	Predefine S32G3XX as a macro, with definition 1
-DS32R45	Predefine S32R45 as a macro, with definition 1
-DGHS	Predefine GHS as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↵c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT↵RT as a macro, with definition 1. Allows drivers to be configured in user mode

3.1.2.2 GHS Assembler Options

Assembler Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-preprocess_assembly_files	Controls whether assembly files with standard extensions such as .s and .asm are preprocessed

Assembler Option	Description
-list	Creates a listing by using the name and directory of the object file with the .lst extension
-c	Stop after assembly and produce an object file for each source file

3.1.2.3 GHS Linker Options

Linker Option	Description
-e Reset_Handler	Make the symbol Reset_Handler be treated as a root symbol and the start label of the application
-T linker_script_file.ld	Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)
-map	Produce a map file
-keepmap	Controls the retention of the map file in the event of a link error
-Mn	Generates a listing of symbols sorted alphabetically/numerically by address
-delete	Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them
-ignore_debug_references	Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers
-Llibrary_path	Points to library_path (the libraries location) for thumb2 to be used for linking
-larch	Link architecture specific library
-lstartup	Link run-time environment startup routines. The source code for themodules in this library is provided in the src/libstartup directory
-lind_sd	Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library
-v	Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols
-nostartfiles	Controls the start files to be linked into the executable

3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the FR driver for S32 microcontrollers. To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

Flexray Files

- ..\Fr_43_LLCE_TS_T40D11M10I3R0\src\Fr_43_LLCE.c
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\src\Fr_43_LLCE_Ipw.c
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\src\Flexray_43_LLCE_Ip.c

Building the driver

- ..\Fr_43_LLCE_TS_T40D11M10I3R0\src\Flexray_43_LLCE_Ip_Irq.c
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Fr_43_LLCE.h
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Fr_43_LLCE_Ipw.h
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Flexray_43_LLCE_Ip.h
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Flexray_43_LLCE_Ip_Irq.h
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Flexray_43_LLCE_Ip_Types.h
- ..\Fr_43_LLCE_TS_T40D11M10I3R0\include\Flexray_43_LLCE_Ip_HwAccess.h

Flexray Generated Files (must be generated by the user using a configuration tool)

- Fr_43_LLCE_Cfg.h
- Flexray_43_LLCE_Ip_Cfg.h
- Fr_43_LLCE_[VariantName]_PBcfg.h
- Fr_43_LLCE_[VariantName]_PBcfg.c
- Flexray_43_LLCE_Ip_[VariantName]_PBcfg.h
- Flexray_43_LLCE_Ip_[VariantName]_PBcfg.c

As a deviation from standard:

- Fr_43_LLCE_Cfg.h - For driver compilation, this file should be generated by the user using a configuration tool.
- Fr_43_LLCE_[VariantName]_PBcfg.h - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the export of the init pointer for the respective variant. This file alone does not contain the whole structure needed by FR_Init function to configure the driver. Based on the number of variants configured in the EcuC, there can be more than one configuration structure for one module even for PreCompile variant.
- Fr_43_LLCE_[VariantName]_PBcfg.c files will contain the definition for all parameters that are variant aware, independent of the configuration class that will be selected (PC, PB). For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.

Files from BaseNXP common folder

- ..\BaseNXP_TS_T40D11M30I0R0\include\
- ..\BaseNXP_TS_T40D11M30I0R0\header\
- ..\BaseNXP_TS_T40D11M30I0R0\src\OsIf_Timer.c
- ..\BaseNXP_TS_T40D11M30I0R0\src\OsIf_Timer_System.c

Files from Dem folder:

- ..\Dem_TS_T40D11M30I0R0\include\Dem.
- ..\Dem_TS_T40D11M30I0R0\src\Dem.c

Files from Det folder:

- ..\Det_TS_T40D11M30I0R0\include\Det.
- ..\Det_TS_T40D11M30I0R0\src\Det.c

Note: <plugin_name>: TS_T<40>D<11>M<SW_Version_Major>I<SW_Version_Minor>R0 (i.e. Target_Id = 2 identifies PowerPC architecture and Derivative_Id = 11 identifies the S32)

3.3 Setting up the plugins

The Flexray driver was designed to be configured by using the EB Tresos Studio (version 27.1.0 b200625-0900 or later or later.)

Location of various files inside the Fr_43_LLCE module folder:

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\config\Fr_43_LLCE.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
 - Fr_43_LLCE_TS_T40D11M10I3R0\autosar\Fr_43_LLCE.epd
- Code Generation Templates for parameters without variation points:
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PC\include\Fr_43_LLCE_Cfg.h
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PC\include\Flexray_43_LLCE_Ip_Cfg.h
- Code Generation Templates for variant aware parameters:
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PC\include\Fr_43_LLCE_PBcfg.h
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PB\src\Fr_43_LLCE_PBcfg.c
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PC\include\Flexray_43_LLCE_Ip_PBcfg.h
 - ..\Fr_43_LLCE_TS_T40D11M10I3R0\generate_PB\src\Flexray_43_LLCE_Ip_PBcfg.c

Steps to generate the configuration:

1. Copy the module folders into the Tresos plugins folder.
 - Fr_43_LLCE_TS_T40D11M10I3R0
 - FrIf_TS_T40D11M30I0R0
 - BaseNXP_TS_T40D11M30I0R0

Building the driver

- Resource_TS_T40D11M30I0R0
 - Det_TS_T40D11M30I0R0
 - Ecuc_TS_T40D11M30I0R0
 - Rte_TS_T40D11M30I0R0
 - Os_TS_T40D11M30I0R0
 - Mcu_TS_T40D11M30I0R0
2. Set the desired Tresos Output location folder for the generated sources and header files.
 3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
 4. Generate the configuration files.

Chapter 4

Function calls to module

- [Function Calls during Start-up](#)
- [Function Calls during Shutdown](#)
- [Function Calls during Wake-up](#)

4.1 Function Calls during Start-up

This driver does not need OS Support. To initialize the FR module, the API `Fr_43_LLCE_Init()` should be called.

4.2 Function Calls during Shutdown

In order to disable FR module, the user can call the `Fr_43_LLCE_DeInit` function. But this function can not apply for platforms which have IpVault lower version 10.

4.3 Function Calls during Wake-up

None.

Chapter 5

Module requirements

- Exclusive areas to be defined in BSW scheduler
- unavailable_exclusive_areas
- Peripheral Hardware Requirements
- ISR to Configure within OS - Dependencies
- ISR Macro
- Other AUTOSAR modules - dependencies
- Data Cache Restrictions
- User Mode support
- Multicore support

5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, FR is using the services of Schedule Manager (SchM) for entering and exiting the exclusive areas. The following critical regions are used in the FR driver:

FR_EXCLUSIVE_AREA_00 is used in function `Fr_43_LLCE_SetAbsoluteTimer` to protect the "FR_TIC↔CR" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ConfigTimer`.

FR_EXCLUSIVE_AREA_01 is used in function `Fr_43_LLCE_SetAbsoluteTimer` to protect the "FR_TIC↔CR" register from read/modify/write operation in `Flexray_43_LLCE_Ip_StartTimer`.

FR_EXCLUSIVE_AREA_02 is used in function `Fr_43_LLCE_ControllerInit` to protect the "FR_TICCR" register from read/modify/write operation in `Flexray_43_LLCE_Ip_StopTimer`.

FR_EXCLUSIVE_AREA_02 is used in function `Fr_43_LLCE_SetAbsoluteTimer` to protect the "FR_TIC↔CR" register from read/modify/write operation in `Flexray_43_LLCE_Ip_StopTimer`.

FR_EXCLUSIVE_AREA_02 is used in function `Fr_43_LLCE_CancelAbsoluteTimer` to protect the "FR_↔TICCR" register from read/modify/write operation in `Flexray_43_LLCE_Ip_StopTimer`.

FR_EXCLUSIVE_AREA_03 is used in function `Fr_43_LLCE_EnableAbsoluteTimerIRQ` to protect the "`FR_PIER0`" and "`FR_PIER1`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_SetProtocolInterrupt`.

FR_EXCLUSIVE_AREA_03 is used in function `Fr_43_LLCE_DisableAbsoluteTimerIRQ` to protect the "`FR_PIER0`" and "`FR_PIER1`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_SetProtocolInterrupt`.

FR_EXCLUSIVE_AREA_04 is used in function `Fr_43_LLCE_EnableAbsoluteTimerIRQ` to protect the "`FR_GIFER`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_SetGlobalInterrupt`.

FR_EXCLUSIVE_AREA_05 is used in function `Fr_43_LLCE_DeInit` to protect the "`FR_GIFER`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ClearGlobalInterruptFlag`.

FR_EXCLUSIVE_AREA_06 is used in function `Fr_43_LLCE_ControllerInit` to protect the "`FR_GIFER`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_Init`.

FR_EXCLUSIVE_AREA_07 is used in function `Fr_43_LLCE_ReceiveRxLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ReceiveInd`.

FR_EXCLUSIVE_AREA_08 is used in function `Fr_43_LLCE_ReceiveRxLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ReceiveInd`.

FR_EXCLUSIVE_AREA_09 is used in function `Fr_43_LLCE_TransmitTxLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_Send`.

FR_EXCLUSIVE_AREA_10 is used in function `Fr_43_LLCE_DisableLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_Abort`.

FR_EXCLUSIVE_AREA_11 is used in function `Fr_43_LLCE_CancelTxLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_CancelTxBuffer`.

FR_EXCLUSIVE_AREA_13 is used in function `Fr_43_LLCE_PreparesLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ReConfigIndividual`.

FR_EXCLUSIVE_AREA_13 is used in function `Fr_43_LLCE_ReconfigLPdu` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ReConfigIndividual`.

FR_EXCLUSIVE_AREA_14 is used in function `Fr_43_LLCE_ControllerInit` to protect the "`FR_MB[index].CCSR`" register from read/modify/write operation in `Flexray_43_LLCE_Ip_ClearDisableIRQs`.

Exclusive Areas implemented in Low level driver layer (IPL)

FR_EXCLUSIVE_AREA_00 is used in function `Flexray_43_LLCE_Ip_ConfigTimer` to protect the updates for:

- `FR_TICCR` register **FR_EXCLUSIVE_AREA_01** is used in function `Flexray_43_LLCE_Ip_StartTimer` to protect the updates for:
- `FR_TICCR` register **FR_EXCLUSIVE_AREA_02** is used in function `Flexray_43_LLCE_Ip_StopTimer` to protect the updates for:
- `FR_TICCR` register **FR_EXCLUSIVE_AREA_03** is used in function `Flexray_43_LLCE_Ip_SetProtocolInterrupt` to protect the updates for:
- `FR_PIER0` register

Module requirements

- FR_PIER1 register **FR_EXCLUSIVE_AREA_04** is used in function Flexray_43_LLCE_Ip_SetGlobalInterrupt to protect the updates for:
- FR_GIFER register **FR_EXCLUSIVE_AREA_05** is used in function Flexray_43_LLCE_Ip_ClearGlobalInterruptFlag to protect the updates for:
- FR_GIFER register **FR_EXCLUSIVE_AREA_06** is used in function Flexray_43_LLCE_Ip_Init to protect the updates for:
- FR_GIFER register **FR_EXCLUSIVE_AREA_07** is used in function Flexray_43_LLCE_Ip_ReceiveInd to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_08** is used in function Flexray_43_LLCE_Ip_ReceiveInd to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_09** is used in function Flexray_43_LLCE_Ip_Send to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_10** is used in function Flexray_43_LLCE_Ip_Abort to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_11** is used in function Flexray_43_LLCE_Ip_CancelTxBuffer to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_12** is used in function Flexray_43_LLCE_Ip_SetBasicDynamicTxFrame to protect the updates for:
- FR_MB[index].CCSR register **FR_EXCLUSIVE_AREA_13** is used in function Flexray_43_LLCE_Ip_ReConfigIndividual to protect the updates for:
- FR_MB[index] register **FR_EXCLUSIVE_AREA_14** is used in function Flexray_43_LLCE_Ip_ReConfigIndividual to protect the updates for:
- FR_MB[index] register

Critical Region Exclusive Matrix

Below is the table depicting the exclusivity between different critical region IDs from the FR driver. If there is an “X” in the table, it means that those 2 critical regions cannot interrupt each other.

The critical regions from interrupts are grouped in “Interrupt Service Routines Critical Regions (composed diagram)”. If an exclusive area is “exclusive” with the composed “Interrupt Service Routines Critical Regions (composed diagram)” group, it means that it is exclusive with each one of the ISR critical regions.

#	FR_00	FR_01	FR_02	FR_03	FR_04	FR_05	FR_06	FR_07	FR_08	FR_09	FR_10	FR_11	FR_12	FR_13	FR_14
FR_00		x	x												

#	F↵ R↵ _↵ E↵ A↵ _00	F↵ R↵ _↵ E↵ A↵ _01	F↵ R↵ _↵ E↵ A↵ _02	F↵ R↵ _↵ E↵ A↵ _03	F↵ R↵ _↵ E↵ A↵ _04	F↵ R↵ _↵ E↵ A↵ _05	F↵ R↵ _↵ E↵ A↵ _06	F↵ R↵ _↵ E↵ A↵ _07	F↵ R↵ _↵ E↵ A↵ _08	F↵ R↵ _↵ E↵ A↵ _09	F↵ R↵ _↵ E↵ A↵ _10	F↵ R↵ _↵ E↵ A↵ _11	F↵ R↵ _↵ E↵ A↵ _12	F↵ R↵ _↵ E↵ A↵ _13	F↵ R↵ _↵ E↵ A↵ _14
F↵ R↵ _↵ E↵ A↵ _01	x		x												
F↵ R↵ _↵ E↵ A↵ _02	x	x													
F↵ R↵ _↵ E↵ A↵ _03															
F↵ R↵ _↵ E↵ A↵ _04						x	x								
F↵ R↵ _↵ E↵ A↵ _05					x		x								
F↵ R↵ _↵ E↵ A↵ _06					x	x		x	x	x	x	x	x	x	x
F↵ R↵ _↵ E↵ A↵ _07							x		x	x	x	x	x	x	x

Module requirements

#	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 00	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 01	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 02	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 03	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 04	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 05	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 06	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 07	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 08	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 09	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 10	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 11	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 12	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 13	F↵ R↵ ↵↵ E↵ A↵ ↵↵ 14
F↵ R↵ ↵↵ E↵ A↵ _08							x	x		x	x	x	x	x	x
F↵ R↵ ↵↵ E↵ A↵ _09							x	x	x		x	x	x	x	x
F↵ R↵ ↵↵ E↵ A↵ _10							x	x	x	x		x	x	x	x
F↵ R↵ ↵↵ E↵ A↵ _11							x	x	x	x	x		x	x	x
F↵ R↵ ↵↵ E↵ A↵ _12							x	x	x	x	x	x		x	x
F↵ R↵ ↵↵ E↵ A↵ _13							x	x	x	x	x	x	x		x
F↵ R↵ ↵↵ E↵ A↵ _14				x	x		x	x	x	x	x	x	x	x	

Note

FR_EA_xx means FR_EXCLUSIVE_AREA_xx

5.2 Peripheral Hardware Requirements

None.

FR_LLCE does not take care of configuring other components but requires that their initialization precede the FR_LLCE initialization. The following components need to be configured:

- External pins - FR_LLCE_A_RX, FR_LLCE_A_TX, FR_LLCE_A_TX_EN, FR_LLCE_B_RX, FR_LLCE_B_TX and FR_LLCE_B_TX_EN pins should be initialized
- The clock for FR_LLCE - Please refer to the chapter Protocol Engine Clocking in S32G Reference Manual [Reference List](#) for the clock configuration (clock source, frequency value)
- FR_LLCE Memory - FR Memory is located in the system memory of the MCU. The FlexRay block does not provide a memory protection scheme for the FR Memory. The FR Memory is a contiguous region where data like, frame headers, physical message buffers payload data, and synchronization tables is stored; the size is maximum 64Kbytes and starts at a 16 byte boundary. It should be configured as cache-inhibited, please see [Data Cache Restriction](#) chapter.

5.3 ISR to Configure within OS - Dependencies

Fr_43_LLCE does not does not generate an interrupt request for each message. In order to monitor the timing of the protocol, AUTOSAR also provides some APIs to process timer interrupt follow:

- Fr_43_LLCE_SetAbsoluteTimer: setup timer API.
- Fr_43_LLCE_EnableAbsoluteTimerIRQ: enable timer interrupt.
- Fr_43_LLCE_GetAbsoluteTimerIRQStatus: get the status of timer interrupt.
- Fr_43_LLCE_AckAbsoluteTimerIRQ: clear the timer interrupt flag.
- Fr_43_LLCE_DisableAbsoluteTimerIRQ: disable timer interrupt.
- Fr_43_LLCE_CancelAbsoluteTimer: disable timer.

On startup, the firmware sets up interrupt routing so that it receives the FR Protocol Interrupt request, and to be able to assert the Host from Core 3 interrupt. When a Protocol Interrupt is generated, the firmware sets bit 0 of HINT3CR, causing IRQ 173 to be triggered on the host (if enabled).

Note

The interrupt does not come directly from the peripheral; instead, it is forwarded via LLCE. The possibility of increased latency should be taken into account.

5.4 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

5.4.1 Without an Operating System The macro `USING_OS_AUTOSAROS` must not be defined.

5.4.1.1 Using Software Vector Mode

The macro `USE_SW_VECTOR_MODE` must be defined and the ISR macro is defined as:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

5.4.1.2 Using Hardware Vector Mode

The macro `USE_SW_VECTOR_MODE` must not be defined and the ISR macro is defined as:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, the drivers' interrupt handlers must also handle the context save and restore.

5.4.2 With an Operating System Please refer to your OS documentation for description of the ISR macro.

5.5 Other AUTOSAR modules - dependencies

- Platform: This module is used for configures platform specific settings, managing the interrupt requests and other system wide settings as defined in each hardware implementation.
- Det: The DET module is used for enabling Development error detection. The API function used is `Det_ReportError()`. The activation / deactivation of Development error detection is configurable using the `McuDevErrorDetect` configuration parameter.
- Dem: This module is necessary for enabling reporting of production relevant error status. The API function used is `Dem_SetEventStatus()`.
- EcuC: The ECUC module is used for ECU configuration. MCAL modules need ECUC to retrieve the variant information.
- Rte: The RTE module is needed for implementing data consistency of exclusive areas that are used by MCU module. The module is the realization (for a particular ECU) of the interfaces of the AUTOSAR Virtual Function Bus (VFB) and thus provides the infrastructure services for communication between Application Software Components as well as facilitating access to basic software components including the OS

- **Base:** The BASE module contains the common files/definitions needed by the MCAL. This means that it is a dependency for all other MCAL modules.
- **Resource:** Resource module is used to select microcontrollers derivatives.
- **Port:** The PORT module is used to configure the Flexray pins.

Base: The Base module contains the common files/definitions needed by all MCAL modules. **Port:** The Port module is used to configure the port pins with the needed modes, before they are used by the FR module. **Mcu:** The Mcu driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required by other MCAL software modules. The clocks need to be initialized prior to using the FR driver. **Det:** The Det module is used for enabling Default error detection. The API function used is Det_ReportError(). The activation / deactivation of Development error detection is configurable using the 'FrDevErrorDetect' configuration parameter. **Dem:** The Dem module is used for enabling reporting of production relevant error status. **EcuC:** The EcuC module is used for ECU configuration. MCAL modules need ECUC to retrieve the variant information. **EcuM:** This module is used for processing the Wakeup notifications of FR. Whenever the module is in 'Sleep' mode and a wakeup event occurs on a wakeup capable channel, it is reported to EcuM through the EcuM_CheckWakeupEvent () API. This is configurable using the 'FrChannelWakeupInfo' configuration parameter. **Rte:** The Rte module is needed for implementing data consistency of exclusive areas that are used by FR module. **Mcl:** The Rte module is needed for implementing CacheClear and CacheFlush operations when data caching is enabled.

5.6 Data Cache Restrictions

None.

To ensure that FR driver always reads latest data updated by the FlexRay CC, FR Memory area has to be configured as cache-inhibited. This is ensured by the Fr_MemoryArea[] array which size and start address corresponds with the FR Memory area and which is located in a non-cacheable section of the MCU system memory.

5.7 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

Currently not supported

5.7.1 User Mode configuration in the module The Fr_43_LLCE module can be run in user mode if the following steps are performed:

- Enable FrEnableUserModeSupport from the configuration
- Call the following functions as trusted functions:

Function syntax	Description	Available via
void Flexray_43_LLCE_Ip_SetUserAccessAllowed(const FLEXRAY_Type *base)	Sets the UAA bit in REG_PROT to make the instance accessible in user mode.	Flexray_43_LLCE_Ip_TrustedFunctions.h
void Flexray_43_LLCE_Ip_ClearUserAccessAllowed(const FLEXRAY_Type *base)	Clears the UAA bit in REG_PROT to make the instance accessible in user mode.	Flexray_43_LLCE_Ip_TrustedFunctions.h

5.7.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may have the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_Ip_TrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to be visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_Ip_TrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

See the sequence chart below for an example calling `Linflexd_Uart_Ip_Init_Privileged()` as a trusted function.

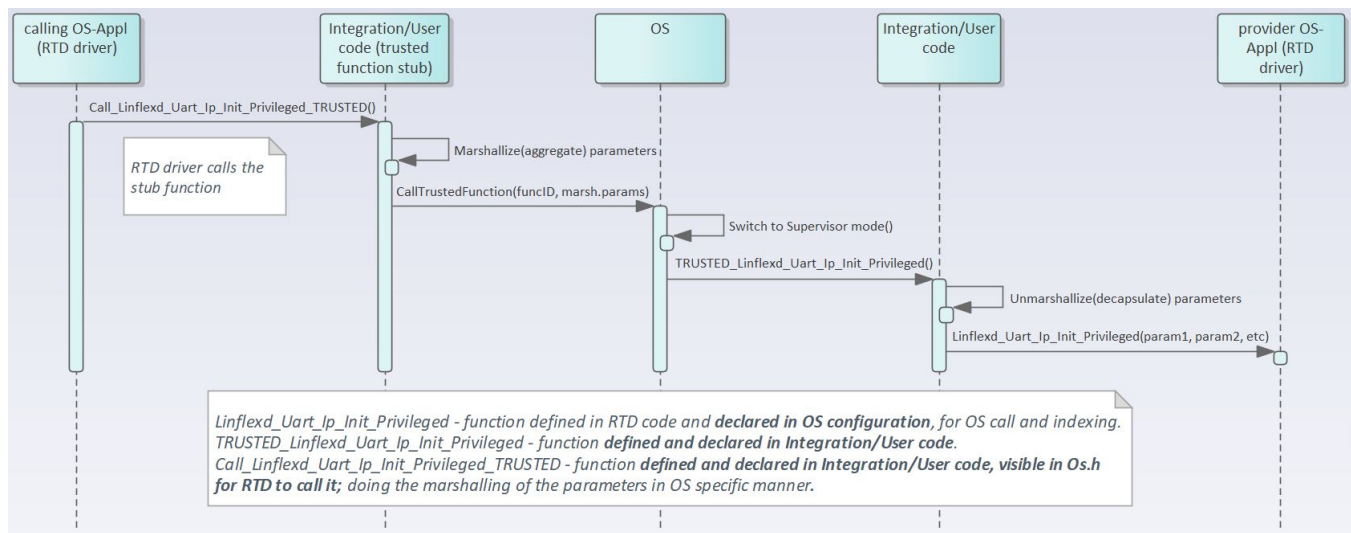


Figure 5.1 Example sequence chart for calling `Linflexd_Uart_Ip_Init_Privileged` as trusted function

5.8 Multicore support

1. The Fr implements the "Autosar 4.4 MCAL Multicore Distribution" according to type II, in which the mappable element is set to HW Unit. For additional details, please refer to AUTOSAR_EXP_BSWDistributionGuide.
2. The Fr and the mappable elements can be allocated to zero, one or several ECUC partitions, by means of "FrEcucPartionRef". If the Fr is mapped to zero ECUC partitions, the Fr behavior reverts to single-core implementation, similar to previous Autosar versions. If the Fr is mapped to one or more ECUC partitions, the Fr enforces the following multi-core assumptions:

The Fr assumes there is a single EcucPartition allocated per core. Internally, the module will use the Core ID returned by Sys_GetCoreID API to reference the appropriate global data and configuration elements. The Fr assumes the EcucCoreIDs are defined in a compact/consecutive order, starting from zero. The rationale is that the number of EcucPartitions is used for dimensioning the Fr internal variables and the EcucCoreIDs are used for indexing those variables. The Fr assumes that initialization is performed on each core, Fr_43_LLCE_Init() is called separately for each core, using a different configuration structure. (Type II) The Fr initialization expects the upper layer will pass the correct initialization pointer, specific to the partition in which the driver is to be used. For example: EcucPartition_1 is assigned to CoreID 1; Fr_43_LLCE_Init function will be called with Fr_Config_↵ EcucPartition_1 configuration structure, on Core 1. The Fr will check upon each API call if the requested resource is configured to be available on the current core, if DET error reporting is enabled. The Fr requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multicore context. The Fr assumes that RTE module implements the EXCLUSIVE AREAS to be core-aware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different cores already, thus implementing the EXCLUSIVE AREAS in a blocking manner (ex: spin-lock) on a multicore scope, might affect the performance of the drivers on the two cores, although they might access separate HW elements. For single-core scope, the EXCLUSIVE AREAS keep the same purpose as on previous AUTOSAR implementations. (* - to be updated per Fr usecase, to be detailed/removed if some modules require such kind of functionality for critical features which cannot be atomically shared among cores). The Fr assumes that each interrupt is routed by the system only to the core on which is supposed to be serviced.



Chapter 6

Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API Requirements](#)
- [Calls to Notification Functions, Callbacks, Callouts](#)

6.1 Main function calls within BSW scheduler

None.

6.2 API Requirements

None.

6.3 Calls to Notification Functions, Callbacks, Callouts

None.

Chapter 7

Memory allocation

- [Sections to be defined in Fr_43_LLCE_MemMap.h](#)
- [Linker command file](#)

7.1 Sections to be defined in Fr_43_LLCE_MemMap.h

Section name	Type of section	Description
FR_43_LLCE_START_SEC_CONFIG_DATA_UNSPECIFIED↔	Configuration Data	Start of Memory Section for Config Data.
FR_43_LLCE_STOP_SEC_CONFIG_DATA_UNSPECIFIED↔	Configuration Data	End of Memory Section for Config Data.
FR_43_LLCE_START_SEC_CODE	Code	Start of memory Section for Code.
FR_43_LLCE_STOP_SEC_CODE	Code	End of memory Section for Code.
FR_43_LLCE_START_SEC_VARIABLES_CLEARED_BOOLEAN↔	Variables	Used for variables which have to be aligned to boolean type. These variables are cleared to zero by start-up code.
FR_43_LLCE_STOP_SEC_VARIABLES_CLEARED_BOOLEAN↔	Variables	End of above section.
FR_43_LLCE_START_SEC_VARIABLES_CLEARED_UNSPECIFIED_NO_CACHEABLE↔	Non-Cacheable Variables	Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit, and that have to be stored in a non-cacheable memory section. These variables are cleared to zero by start-up code.
FR_43_LLCE_STOP_SEC_VARIABLES_CLEARED_UNSPECIFIED_NO_CACHEABLE↔	Non-Cacheable Variables	End of above section.
FR_43_LLCE_START_SEC_VARIABLES_CLEARED_8↔	Variables	Used for variables which have to be aligned to 8 bit. For instance used for variables of size 8 bit or used for composite data types: arrays, structs containing elements of maximum 8 bits. These variables are cleared to zero by start-up code.

Section name	Type of section	Description
FR_43_LLCE_STOP_SEC_VAR_↵ CLEARED_8	Variables	End of above section.
FR_43_LLCE_START_SEC_VAR_↵ _CLEARED_UNSPECIFIED	Variables	Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are cleared to zero by start-up code.
FR_43_LLCE_STOP_SEC_VAR_↵ CLEARED_UNSPECIFIED	Variables	End of above section.
FR_43_LLCE_START_SEC_VAR_↵ _INIT_BOOLEAN	Variables	Used for variables which have to be aligned to boolean type. These variables are initialized with values after every reset.
FR_43_LLCE_STOP_SEC_VAR_I↵ _INIT_BOOLEAN	Variables	End of above section.
FR_43_LLCE_START_SEC_CONST↵ _UNSPECIFIED	Constant Data	Used for constants that does not fit the criteria of 8,16 or 32 bit.
FR_43_LLCE_STOP_SEC_CONST↵ _UNSPECIFIED	Constant Data	End of above section.

7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>_MemMap.h.

Chapter 8

Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

Chapter 9

External assumptions for driver

The section presents requirements that must be complied with when integrating the FR_LLCE driver into the application.

External Assumption Req ID	External Assumption Text
SWS_Fr_00117	Fr_GeneralTypes.h shall contain all types and constants that are shared among the AUTOSAR FlexRay modules Fr, FrIf and FrTrcv. Note: Must be ensured by integrator. FR driver does not provide this file.
SWS_Fr_00499	The content of Fr_GeneralTypes.h shall be protected by a FR_GENERAL_TYPES define. Note: Must be ensured by integrator.
SWS_Fr_00500	If different FlexRay drivers are used, only one instance of this file has to be included in the source tree. For implementation all Fr_GeneralTypes.h related types in the documents mentioned before shall be considered. Note: Must be ensured by integrator.
SWS_Fr_00498	Error Name: - FR_E_CTRL_TESTRESULT [_<Fr_CtrlIdx>] - Short Description: - FlexRay Controller hardware error - Long Description: - This extended production error indicates hardware errors of the FlexRay communication controller. Please note that this extended production error does not address Flexray protocol errors detected on the network. - Detection Criteria: - Fail - Every API function accessing hardware registers of the FlexRay controller might detect a missbehavior of the device compared to the device specification. For details see requirements: SWS_Fr_00147, SWS_Fr_00176, SWS_Fr_00181, SWS_Fr_00520, SWS_Fr_00186, SWS_Fr_00190, SWS_Fr_00195, SWS_Fr_00201, SWS_Fr_00216, SWS_Fr_00223, SWS_Fr_00613, SWS_Fr_00232, SWS_Fr_00243, SWS_Fr_00248, SWS_Fr_00529, SWS_Fr_00543, SWS_Fr_00255, SWS_Fr_00261, SWS_Fr_00552, SWS_Fr_00560, SWS_Fr_00568, SWS_Fr_00580, SWS_Fr_00589, SWS_Fr_00272, SWS_Fr_00286, SWS_Fr_00297, SWS_Fr_00308, SWS_Fr_00319, SWS_Fr_00331, SWS_Fr_00652 - Pass - During FlexRay communication controller initialization (function Fr_ControllerInit()) the proper operation of the hardware registers is successfully verified. For details see requirements: SWS_Fr_00598, - Secondary Parameters: - In case of successful device operation verification (function Fr_ControllerInit()) report PASS. In case of an error always report FAIL. - Time Required: - If a FlexRay Controller hardware error occurs it shall be immediately reported as error. - Monitor Frequency - continuous - Note: Value for this production code Event Id is assigned externally by the configuration of the Dem.

External Assumption Req ID	External Assumption Text
SWS_Fr_00600	Error Name: - FR_E_LPDU_SLOTSTATUS [_{<}LPduIdx_{>}] - Short Description: - FlexRay Protocol communication error - Long Description: - This production error indicates Flexray protocol communication errors on the network for a particular LPdu. - Detection Criteria: - Fail - Each time FlexRay slot status with at least one of the following FlexRay protocol errors active:vSS!SyntaxErrorvSS!ContentErrorvSS!BviolationFor details see requirements:SWS_Fr_00605, SWS_Fr_00606, - Pass - Each time FlexRay slot status with none one of the following FlexRay protocol errors active:vSS!SyntaxErrorvSS!ContentErrorvSS!BviolationFor details see requirements:SWS_Fr_00627, SWS_Fr_00629, - Secondary Parameters: - Detection of production error events is active only during ongoing FlexRay communication. - Time Required: - The time for detecting an evident FlexRay protocol error in a particular slot strongly depends on the period of the actual FlexRay slot and thus on the FlexRay protocol configuration parameters. - Monitor Frequency - continuous - Note: Value for this production code Event Id is assigned externally by the configuration of the Dem.
SWS_Fr_00505	Name: - Fr_POCStateType - Type: - Enumeration - Range: - FR_POCSTATE_CONFIG - 0x00 - Represents literal CONFIG of formal type definition T_POCState. - FR_POCSTATE_DEFAULT_CONFIG - 0x01 - Represents literal DEFAULT_CONFIG of formal type definition T_POCState. - FR_POCSTATE_HALT - 0x02 - Represents literal HALT of formal type definition T_POCState. - FR_POCSTATE_NORMAL_ACTIVE - 0x03 - Represents literal NORMAL_ACTIVE of formal type definition T_POCState. - FR_POCSTATE_NORMAL_PASSIVE - 0x04 - Represents literal NORMAL_PASSIVE of formal type definition T_POCState. - FR_POCSTATE_READY - 0x05 - Represents literal READY of formal type definition T_POCState. - FR_POCSTATE_STARTUP - 0x06 - Represents literal STARTUP of formal type definition T_POCState. - FR_POCSTATE_WAKEUP - 0x07 - Represents literal WAKEUP of formal type definition T_POCState. - Description: - This formal definition refers to the description of type T_POCState in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h - Note: Fr_POCStateType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00506	Name: - Fr_SlotModeType - Type: - Enumeration - Range: - FR_SLOTMODE_KEY_SLOT - 0x00 - Represents literal KEY_SLOT of formal type definition T_SlotMode. - FR_SLOTMODE_ALL_PENDING - 0x01 - Represents literal ALL_PENDING of formal type definition T_SlotMode. - FR_SLOTMODE_ALL - 0x02 - Represents literal ALL of formal type definition T_SlotMode. - Description: - This formal definition refers to the description of type T_SlotMode in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h - Note: Fr_SlotModeType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00507	Name: - Fr_ErrorModeType - Type: - Enumeration - Range: - FR_ERRORMODE_ACTIVE - 0x00 - Represents literal ACTIVE of formal type definition T_ErrorMode. - FR_ERRORMODE_PASSIVE - 0x01 - Represents literal PASSIVE of formal type definition T_ErrorMode. - FR_ERRORMODE_COMM_HALT - 0x02 - Represents literal COMM_HALT of formal type definition T_ErrorMode. - Description: - This formal definition refers to the description of type T_ErrorMode in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h - Note: Fr_ErrorModeType shall be provided by the Fr_GeneralTypes.h file.

External Assumption Req ID	External Assumption Text
SWS_Fr_00508	Name: - Fr_WakeupStatusType - Type: - Enumeration - Range: - FR_WAKEUP_UNDEFINED - 0x00 - Represents literal UNDEFINED of formal type definition T_WakeupStatus. - FR_WAKEUP_RECEIVED_HEADER - 0x01 - Represents literal RECEIVED_HEADER of formal type definition T_WakeupStatus. - FR_WAKEUP_RECEIVED_WUP - 0x02 - Represents literal RECEIVED_WUP of formal type definition T_WakeupStatus. - FR_WAKEUP_COLLISION_HEADER - 0x03 - Represents literal COLLISION_HEADER of formal type definition T_WakeupStatus. - FR_WAKEUP_COLLISION_WUP - 0x04 - Represents literal COLLISION_WUP of formal type definition T_WakeupStatus. - FR_WAKEUP_COLLISION_UNKNOWN - 0x05 - Represents literal COLLISION_UNKNOWN of formal type definition T_WakeupStatus. - FR_WAKEUP_TRANSMITTED - 0x06 - Represents literal TRANSMITTED of formal type definition T_WakeupStatus. - Description: - This formal definition refers to the description of type T_WakeupStatus in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h - Note: Fr_WakeupStatusType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00509	Name: - Fr_StartupStateType - Type: - Enumeration - Range: - FR_STARTUP_UNDEFINED - 0x00 - Represents literal UNDEFINED of formal type definition T_StartupState. - FR_STARTUP_COLDSTART_LISTEN - 0x01 - Represents literal COLDSTART_LISTEN of formal type definition T_StartupState. - FR_STARTUP_INTEGRATION_COLDSTART_CHECK - 0x02 - Represents literal INTEGRATION_COLDSTART_CHECK of formal type definition T_StartupState. - FR_STARTUP_COLDSTART_JOIN - 0x03 - Represents literal COLDSTART_JOIN of formal type definition T_StartupState. - FR_STARTUP_COLDSTART_COLLISION_RESOLUTION - 0x04 - Represents literal COLDSTART_COLLISION_RESOLUTION of formal type definition T_StartupState. - FR_STARTUP_COLDSTART_CONSISTENCY_CHECK - 0x05 - Represents literal COLDSTART_CONSISTENCY_CHECK of formal type definition T_StartupState. - FR_STARTUP_INTEGRATION_LISTEN - 0x06 - Represents literal INTEGRATION_LISTEN of formal type definition T_StartupState. - FR_STARTUP_INITIALIZE_SCHEDULE - 0x07 - Represents literal INITIALIZE_SCHEDULE of formal type definition T_StartupState. - FR_STARTUP_INTEGRATION_CONSISTENCY_CHECK - 0x08 - Represents literal INTEGRATION_CONSISTENCY_CHECK of formal type definition T_StartupState. - FR_STARTUP_COLDSTART_GAP - 0x09 - Represents literal COLDSTART_GAP of formal type definition T_StartupState. - FR_STARTUP_EXTERNAL_STARTUP - 0x0a - Represents literal EXTERNAL_STARTUP of formal type definition T_StartupState. - Description: - This formal definition refers to the description of type T_StartupState in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h
SWS_Fr_00510	Name: - Fr_POCStatusType - Type: - Structure - Element: - boolean - CHIHaltRequest - - - boolean - ColdstartNoise - - - Fr_ErrorModeType - ErrorMode - - - boolean - Freeze - - - Fr_SlotModeType - SlotMode - - - Fr_StartupStateType - StartupState - - - Fr_POCStateType - State - - - Fr_WakeupStatusType - WakeupStatus - - - boolean - CHIReadyRequest - - - Description: - This formal definition refers to the description of type T_POCStatus in chapter 2.2.1.3 POC status of [12]. - Available via: - Fr_GeneralTypes.h - Note: Fr_POCStatusType shall be provided by the Fr_GeneralTypes.h file.

External Assumption Req ID	External Assumption Text
SWS_Fr_00511	Name: - Fr_TxLPduStatusType - Type: - Enumeration - Range: - FR_TRANSMITTED - 0x00 - LPdu has been transmitted - FR_TRANSMITTED_CONFLICT - 0x01 - A transmission conflict has occurred. - FR_NOT_TRANSMITTED - 0x02 - LPdu has not been transmitted - Description: - These values are used to determine whether a LPdu has been transmitted or not. - Available via: - Fr_GeneralTypes.h - Note: Fr_TxLPduStatusType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00512	Name: - Fr_RxLPduStatusType - Type: - Enumeration - Range: - FR_RECEIVED - 0x00 - LPdu has been received - FR_NOT_RECEIVED - 0x01 - LPdu has not been received - FR_RECEIVED_MORE_DATA_AVAILABLE - 0x02 - LPdu has been received. More instances of this LPdu are available (FIFO usage). - Description: - These values are used to determine if a LPdu has been received or not. - Available via: - Fr_GeneralTypes.h - Note: Fr_RxLPduStatusType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00514	Name: - Fr_ChannelType - Type: - Enumeration - Range: - FR_CHANNEL_A - 0x01 - Refers to channel A of a CC. - FR_CHANNEL_B - 0x02 - Refers to channel B of a CC. - FR_CHANNEL_AB - 0x03 - Refers to both channels (A and B) of a CC. - Description: - The values are used to reference channels on a CC. - Available via: - Fr_GeneralTypes.h - Note: Fr_ChannelType shall be provided by the Fr_GeneralTypes.h file.
SWS_Fr_00390	API function - Header File - Description - Dem_SetEventStatus - Dem.h - Called by SW-Cs or BSW modules to report monitor status information to the Dem. BSW modules calling Dem_SetEventStatus can safely ignore the return value. - Det_ReportRuntimeError - Det.h - Service to report runtime errors. If a callout has been configured then this callout shall be called. -
SWS_Fr_00391	API function - Header File - Description - Det_ReportError - Det.h - Service to report development errors. - SchM_Enter_Fr - - - SchM_Exit_Fr - - - Note: This requirement defines interfaces that are required to fulfill an optional functionality of the module. SchM functionality is not used by the FR driver.
EA_RTD_00071	If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used.
EA_RTD_00081	The integrator shall assure that <MSN>_Init() and <MSN>_DeInit() functions do not interrupt each other.
EA_RTD_00082	When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: Rationale: This ensures that no other buffers/variables compete for the same cache lines.
EA_RTD_00092	The integrator shall allocate a single EcucPartition per core or the partition in which the Fr_llce is allocated shall be exclusively mapped to a core. Note: Internally, the Fr_llce will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements, that is why a core should reference only one configured partition.
EA_RTD_00093	The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero.
EA_RTD_00094	When multicore support is enabled, the application shall call Fr_llce_Init() for each core, using the dedicated configuration pointer for that core.

External assumptions for driver

External Assumption Req ID	External Assumption Text
EA_RTD_00096	The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used.
EA_RTD_00113	When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention Call<Function_Name>TRUSTED(parameter1,parameter2,...) in Integration/User code. They need to be visible in Os.h for the driver to call them. They will call RTD <Function_Name>() as trusted functions in OS specific manner.

How to Reach Us:

Home Page:

nxp.com

Web Support:

nxp.com/support

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals," must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address: nxp.com/SalesTermsandConditions.

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, COOLFLUX, EMBRACE, GREENCHIP, HITAG, I2C BUS, ICODE, JCOP, LIFE VIBES, MIFARE, MIFARE CLASSIC, MIFARE DESFire, MIFARE PLUS, MIFARE FLEX, MANTIS, MIFARE ULTRALIGHT, MIFARE4MOBILE, MIGLO, NTAG, ROADLINK, SMARTLX, SMARTMX, STARPLUG, TOPFET, TRENCHMOS, UCODE, Freescale, the Freescale logo, Altivec, C-5, CodeTEST, CodeWarrior, ColdFire, ColdFire+, C-Ware, the Energy Efficient Solutions logo, Kinetis, Layerscape, MagniV, mobileGT, PEG, PowerQUICC, Processor Expert, QorIQ, QorIQ Qonverge, Ready Play, SafeAssure, the SafeAssure logo, StarCore, Symphony, VortiQa, Vybrid, Airfast, BeeKit, BeeStack, CoreNet, Flexis, MXC, Platform in a Package, QUICC Engine, SMARTMOS, Tower, TurboLink, and UMEMS are trademarks of NXP B.V. All other product or service names are the property of their respective owners. ARM, AMBA, ARM Powered, Artisan, Cortex, Jazelle, Keil, SecurCore, Thumb, TrustZone, and Vision are registered trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. ARM7, ARM9, ARM11, big.LITTLE, CoreLink, CoreSight, DesignStart, Mali, mbed, NEON, POP, Sensinode, Socrates, ULINK and Versatile are trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. All rights reserved. Oracle and Java are registered trademarks of Oracle and/or its affiliates. The Power Architecture and Power.org word marks and the Power and Power.org logos and related marks are trademarks and service marks licensed by Power.org.

© 2023 NXP B.V.

